

Rでの解析の流れ

```
> l8-8A_seiseki <- read.table('./seiseki_zenkikouki.csv', header=T, sep=',')
> head(l8-8A_seiseki)
  class no kokugo1 kokugo2 seikei1 seikei2 sugaku1 sugaku2 butsuril butsuril2
1     1  1      58      51      67      52      69      28      69      24
2     1  2      55      47      66      35      74      69      78      43
3     1  3      58      58      83      57      53      37      78      34
4     1  4      37      42      60      46      60      16      81      22
5     1  5      69      61      60      21      45      50      90      40
6     1  6      63      63      61      34      28      52      69      34

  eigol eigo2
1     82     62
2     72     53
3     53     44
4     31     11
5     40     27
6     85     67

> l8-8A_seiseki_fact <- factanal(l8-8A_seiseki[,3:12], factors=3, scores='Bartlett', rotation='varimax')
> l8-8A_seiseki_fact

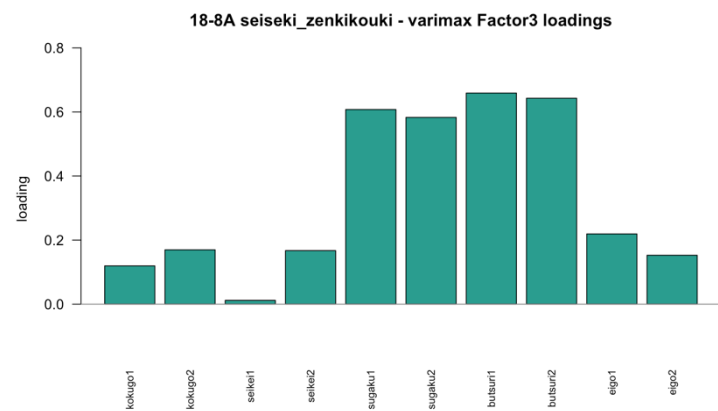
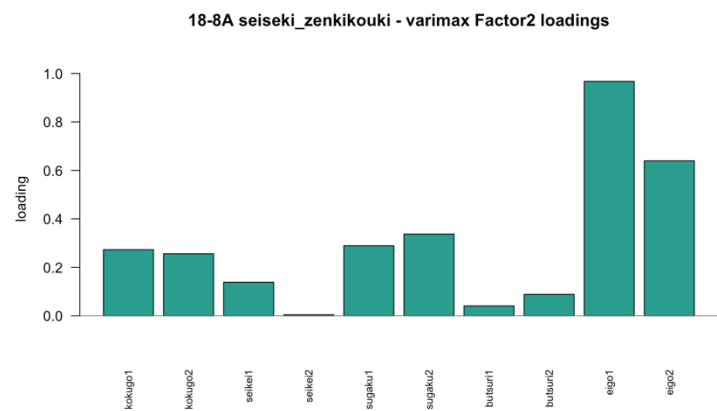
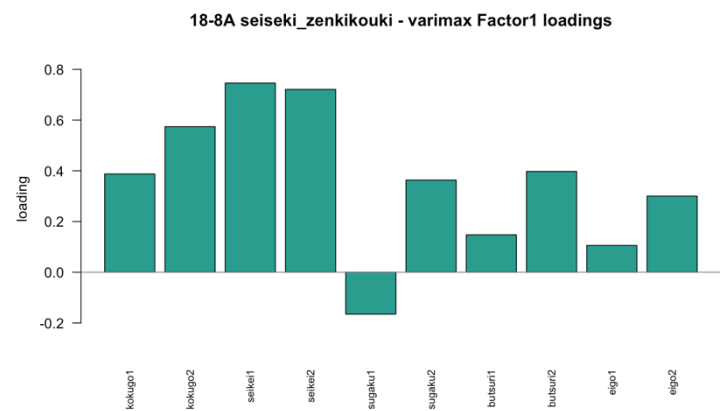
Call:
factanal(x = df[, data_cols], factors = factors, scores = scores, rotation = rotation)

Uniquenesses:
  kokugo1 kokugo2 seikei1 seikei2 sugaku1 sugaku2 butsuril butsuril2
    0.761   0.576   0.424   0.452   0.520   0.414   0.542   0.421
    eigol   eigo2
    0.005   0.477

Loadings:
      Factor1 Factor2 Factor3
kokugo1  0.388  0.273  0.120
kokugo2  0.574  0.256  0.170
seikei1   0.746  0.139      0
seikei2   0.721      0  0.167
sugaku1  -0.165  0.289  0.608
sugaku2   0.364  0.337  0.583
butsuril  0.148      0  0.659
butsuril2 0.397      0  0.643
eigol     0.106  0.967  0.219
eigo2     0.300  0.640  0.153

      Factor1 Factor2 Factor3
```

因子負荷量の棒グラフ



ソースコード (1/2)

Class13_rp18-8A_02.R

```
1 setwd("/Users/kakigoori/Desktop/data/13")
2
3 dir.create("./answers/plots_pdf", recursive=TRUE, showWarnings=FALSE)
4
5 seiseki <- read.table("./seiseki_zenkikouki.csv", header=T, sep=",")
6
7 seiseki
8
9 head(seiseki, 5)
10
11 seiseki_fact <- factanal(seiseki[,3:12], factors=3)
12
13 seiseki_fact
14
15 names(seiseki_fact)
16
17 seiseki_fact <- factanal(seiseki[,3:12], factors=3, scores="Bartlett")
18
19 names(seiseki_fact)
20
21 head(seiseki_fact$scores, 5)
22
23 tail(seiseki_fact$scores, 5)
24
```

ソースコード (2/2)

Class13_rp18-8A_02.R

```
25 seiseki_fact$loadings[,1:3]
26
27 pdf("./answers/plots_pdf/18-8A_seiseki_factor1_bar.pdf", width=8, height=5.2)
28 barplot(seiseki_fact$loadings[,1], cex.names=0.8, las=2, ylim=c(-0.3,1))
29 abline(h=0, col="gray50")
30 dev.off()
31
32 pdf("./answers/plots_pdf/18-8A_seiseki_factor2_bar.pdf", width=8, height=5.2)
33 barplot(seiseki_fact$loadings[,2], cex.names=0.8, las=2, ylim=c(-0.3,1))
34 abline(h=0, col="gray50")
35 dev.off()
36
37 pdf("./answers/plots_pdf/18-8A_seiseki_factor3_bar.pdf", width=8, height=5.2)
38 barplot(seiseki_fact$loadings[,3], cex.names=0.8, las=2, ylim=c(-0.3,1))
39 abline(h=0, col="gray50")
40 dev.off()
```